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Working with Services

In our previous section, we discussed filtering and the pipeline using an example of finding services on the host from the eyes of a pentester. In this section, we will dive deeper into this and flip it on its head. We are going to look at it from the eyes of an administrator.

Scenario: Mr. Tanaka messaged the Helpdesk stating that he noticed a window pop up earlier in the day and thought it was just Windows updates running, as lots of information flashed by in the window. However, now he reports that alerts stating that Defender is turned off also popped up, and his host is acting sluggish. We need to look into this, determine what services related to Defender are shut off, and enable them again if we can. Later we will look into the event logs and see what happened.

Service administration is crucial in managing hosts and ensuring our security posture remains unchanged. This section will cover how to query, start, stop, and edit services and their permissions as needed. We will also discuss ways to interact with them locally and remotely. It is time to dive in and acquire our next CLI Kung-Fu Belt.

What Are Services and How Do We Interact with Them Using Powershell?

Services in the Windows Operating system at their core are singular instances of a component running in the background that manages and maintains processes and other needed components for applications used on the host. Services usually do not require interaction from the user and have no tangible interface for them to interact with. They also exist as a singular instance of the service on the host, while a service can maintain multiple instances of a process. A process can be considered a temporary container for a user or application to perform tasks. Windows has three categories of services: Local Services, Network Services, and System Services. Many different services (including the core components within the Windows operating system) handle multiple instances of processes simultaneously. PowerShell provides us with the module `Microsoft.PowerShell.Management`, which contains several cmdlets for interacting with Services. As with everything in PowerShell, if you are unsure where to start or what cmdlet you need, take advantage of the built-in help to get you started.

Getting Help (Services)

Working with Services			
PS C:\htb> Get-Help *-Service			
Name	Category	Module	Synopsis
----	-----	-----	-----
Get-Service	Cmdlet	Microsoft.PowerShell.Man...	...
New-Service	Cmdlet	Microsoft.PowerShell.Man...	...
Remove-Service	Cmdlet	Microsoft.PowerShell.Man...	...
Restart-Service	Cmdlet	Microsoft.PowerShell.Man...	...
Resume-Service	Cmdlet	Microsoft.PowerShell.Man...	...
Set-Service	Cmdlet	Microsoft.PowerShell.Man...	...
Start-Service	Cmdlet	Microsoft.PowerShell.Man...	...
Stop-Service	Cmdlet	Microsoft.PowerShell.Man...	...
Suspend-Service	Cmdlet	Microsoft.PowerShell.Man...	...

Now, let us start our triage of Mr. Tanaka's host and see what is going on.



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Investigating Running Services

We first need to get a quick running list of services from our target host. Services can have a status set as Running, Stopped, or Paused and can be set up to start manually (user interaction), automatically (at system startup), or on a delay after system boot. Users with administrative privileges can usually create, modify, and delete services. Misconfigurations around service permissions are a common privilege escalation vector on Windows systems.

Get-Service

Working with Services

```
PS C:\htb> Get-Service | ft DisplayName,Status
```

DisplayName	Status
-----	-----
Adobe Acrobat Update Service	Running
OpenVPN Agent agent_ovpnconnect	Running
Adobe Genuine Monitor Service	Running
Adobe Genuine Software Integrity Service	Running
Application Layer Gateway Service	Stopped
Application Identity	Stopped
Application Information	Running
Application Management	Stopped
App Readiness	Stopped
Microsoft App-V Client	Stopped
AppX Deployment Service (AppXSVC)	Running
AssignedAccessManager Service	Stopped
Windows Audio Endpoint Builder	Running
Windows Audio	Running
ActiveX Installer (AxInstSV)	Stopped
GameDVR and Broadcast User Service_172433	Stopped
BitLocker Drive Encryption Service	Running
Base Filtering Engine	Running
<SNIP>	

```
PS C:\htb> Get-Service | measure
```

```
Count          : 321
```

To make it a little clearer to run, we piped our service listing into `format-table` and chose the properties `DisplayName` and `Status` to display in our console. On the second command issued, we measured the number of services that appear in the listing just to get a sense of how many we are working with. `321` services are a lot to scroll through and work with at once, so we need to pare it down a bit more. From Mr. Tanaka's request, he mentioned a potential issue with Windows Defender, so let us filter out any services not related to that.

Precision Look at Defender

Working with Services

```
PS C:\htb> Get-Service | where DisplayName -like '*Defender*' | ft DisplayName,ServiceName,Status
```

DisplayName	ServiceName	Status
-----	-----	-----
Windows Defender	WinDefend	Running



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Now we can see just the services related to **Defender**, and we can see that for some reason, the Microsoft Defender Antivirus Service (**WinDefend**) is indeed turned off. For now, to ensure the protection of Mr. Tanaka's host, let us try and turn it back on using the `Start-Service` cmdlet.

Resume / Start / Restart a Service

Working with Services
PS C:\htb> Start-Service WinDefend

As we ran the cmdlet `Start-Service` as long as we did not get an error message like `"ParserError: This script contains malicious content and has been blocked by your antivirus software."` or others, the command executed successfully. We can check again by querying the service.

Checking Our Work

Working with Services		
PS C:\htb> get-service WinDefend		
Status	Name	DisplayName
-----	----	-----
Running	WinDefend	Microsoft Defender Antivirus Service

Notice how we utilized the service **Name** to start and query the service instead of anything in the **DisplayName**. For now, Defender is back up and running, so the first mission is accomplished. While here, let us look around and see what else is happening. As we look through the services a bit more to see what is there, we notice a Service with an odd **DisplayName**.

Working with Services		
PS C:\htb> get-service		
Stopped	SmsRouter	Microsoft Windows SMS Router Service.
Stopped	SNMPTrap	SNMP Trap
Stopped	spectrum	Windows Perception Service
Running	Spooler	Totally still used for Print Spooli...
Stopped	sppsvc	Software Protection
Running	SSDPsrv	SSDP Discovery

We cannot find any information on this particular service, and its **DisplayName** having changed is odd, so to be safe, we will stop the service for now and let one of our team members on the security team investigate it.

Stopping a Service

Working with Services

PS C:\htb> Stop-Service Spooler

PS C:\htb> Get-Service Spooler

Status	Name	DisplayName
--------	------	-------------



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Now we can see that using the `Stop-Service`, we stopped the operating status of the `spooler` service. Now that we have stopped the service let us set the startup type of the service now from Automatic to Disabled until further investigation can be taken.

Set-Service

Working with Services

```
PS C:\htb> get-service spooler | Select-Object -Property Name, StartType, Status, DisplayName
```

Name	StartType	Status	DisplayName
spooler	Automatic	Stopped	Totally still used for Print Spooling...

```
PS C:\htb> Set-Service -Name Spooler -StartType Disabled
```

```
PS C:\htb> Get-Service -Name Spooler | Select-Object -Property StartType
```

StartType
Disabled

Ok, now our Spooler service has been stopped, and its Startup changed to Disabled for now. Modifying a running service is reasonably straightforward. Ensure that if you attempt to make any modifications, you are an Administrator for the host or on the domain. Removing services in PowerShell is difficult right now. The cmdlet `Remove-Service` only works if you are using PowerShell version 7. By default, our hosts will open and run PowerShell version 5.1. For now, if you wish to remove a service and its entries, use the `sc.exe` tool.

How Do We Interact with Remote Services using PowerShell?

Now that we know how to work with services, let us look at how we can interact with remote hosts. Since Mr. Tanaka's host is in a domain, we can easily query and check the running services on other hosts. The `-ComputerName` parameter allows us to specify that we want to query a remote host.

Remotely Query Services

Working with Services

```
PS C:\htb> get-service -ComputerName ACADEMY-ICL-DC
```

Status	Name	DisplayName
Running	ADWS	Active Directory Web Services
Stopped	AppIDSvc	Application Identity
Stopped	AppMgmt	Application Management
Stopped	AppReadiness	App Readiness
Stopped	AppXSvc	AppX Deployment Service (AppXSVC)
Running	BFE	Base Filtering Engine
Stopped	BITS	Background Intelligent Transfer Ser...

<SNIP>

Filtering our Output

Working with Services



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Running	AUWS	Active Directory Web Services
Running	BFE	Base Filtering Engine
Running	COMSysApp	COM+ System Application
Running	CoreMessagingRe...	CoreMessaging
Running	CryptSvc	Cryptographic Services
Running	DcomLaunch	DCOM Server Process Launcher
Running	Dfs	DFS Namespace
Running	DFSR	DFS Replication

One interesting thing of note here is that since PowerShell handles everything as an **object**, even the output from a remote command, we can use the PowerShell pipeline to dissect an object's properties with **Where-Object**. Our results returned only the services with a status when it was run of **Running**. We can use these combinations for any number of things. One great example would be to query our hosts for a specific property, such as if the status was Running, if a DisplayName is set to something specific, etc. Regarding remote interactions, we can also use the **Invoke-Command** cmdlet. Let us try and query multiple hosts and see the status of the **UserManager** service.

Invoke-Command

Working with Services

```
PS C:\htb> invoke-command -ComputerName ACADEMY-ICL-DC,LOCALHOST -ScriptBlock {Get-Service -Name 'windefend'}
```

Status	Name	DisplayName	PSComputerName
----	----	-----	-----
Running	windefend	Microsoft Defender Antivirus Service	LOCALHOST
Running	windefend	Windows Defender Antivirus Service	ACADEMY-ICL-DC

Let us break this down now:

- **Invoke-Command**: We are telling PowerShell that we want to run a command on a local or remote computer.
- **Computername**: We provide a comma-defined list of computer names to query.
- **ScriptBlock {commands to run}**: This portion is the enclosed command we want to run on the computer. For it to run, we need it to be enclosed in {}.

Interacting with hosts in this manner can expedite much of our work.

Scenario: Earlier in this section, we saw a service (Spooler) that had a DisplayName that was modified. This could potentially clue us in on an issue within our environment. Using the **-ComputerName** parameter or the Invoke-Command cmdlet to query all of the hosts within our environment and check the DisplayName properties to see if any other host has been affected. As an administrator, having access to this kind of power is invaluable and can often help reduce the time a threat is on the host and help get ahead of the issue and work to kick the threat out.

Understanding services and managing them on a host is essential from an admin and pentester perspective. We can do a lot with them, including anything from privilege escalation, to persistence and more. Moving on to our next section, we will introduce the Windows Registry and how to interact with it on a Windows host.

VPN Servers

Warning: Each time you "Switch", your connection keys are regenerated and you must re-download your VPN connection file.



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PROTOCOL

☒ UDP 1337 ☐ TCP 443

DOWNLOAD VPN CONNECTION FILE



Connect to Pwnbox

Your own web-based Parrot Linux instance to play our labs.

Pwnbox Location

AU

100000ms ▼

Terminate Pwnbox to switch location

Start Instance

∞ / 1 spawns left

Waiting to start...



Questions

Answer the question(s) below to complete this Section and earn cubes!

Target(s): [Click here to spawn the target system!](#)

+ 0

What Cmdlet will show us the current services of a host?

Submit your answer here...



Cheat Sheet



Download VPN
Connection File



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+ 0

If we wanted to start the Windows Defender Service, what command would we use?

Submit your answer here...

+10 Streak pts

Submit

Hint

Show Solution

+ 0

What Cmdlet will allow us to execute a command on a remote host?

Submit your answer here...

+10 Streak pts

Submit

Hint

Show Solution

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O F F L I N E

 Start Instance

 / 1 spawns left